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Mapping Technological Knowledge Evolution in Chinese Universities

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Abstract

Introduction. Patent indicators describe inventive output, whereas understanding knowledge evolution also requires tracing sources, technical combinations, emerging directions, and subsequent use. This study develops a framework for Chinese universities.

Method. Dated patent objects connect classifications, text, organizations, and citations through three complementary network layers. Four process dimensions measure acquisition, recombination, recognized emergence, and diffusion. Transformation is assessed independently as sustained structural change.

Analysis. Gaussian mixture models summarize university-period profiles; transition matrices describe state sequences. Fixed citation horizons, university-level resampling, blinded case assessment, and time-separated validation specify how the framework will be evaluated.

Results. This draft provides operational definitions and an empirical reporting design. The source database has not been supplied, so distributions, state counts, and transition probabilities remain uncomputed. Maintenance, expansion, recombination, and transformation are proposed trajectory interpretations rather than established findings.

Conclusion. The framework connects knowledge organization and patent mapping to longitudinal institutional analysis. Its empirical usefulness depends on whether relational distinctions survive controls for patent volume, technological field, coverage, and observation time.

Complete extended working version; empirical analysis pending.

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Introduction

Universities contribute to technological development through inventions that draw on established knowledge, combine technical components, and create opportunities for further work. Patent documents provide a distinctive information resource for studying these processes because they connect technical descriptions to formal classifications, prior-art references, organizational identities, and dates. The analytical challenge is to turn these heterogeneous records into an account of how knowledge structures change. Counting documents alone cannot establish whether an institution is deepening an existing specialization, entering adjacent technical areas, combining previously disconnected elements, or sustaining a new direction.

Patent counts and citations nevertheless have legitimate uses. They support measurement of inventive activity and subsequent technological attention, subject to differences in patenting propensity, sector, jurisdiction, and citation practice (Griliches, 1990; Hall et al., 2005; Nagaoka et al., 2010). Citation-based indicators can also be associated with the value of inventions (Trajtenberg, 1990). The issue addressed here is therefore an additional explanatory requirement: an evolutionary analysis needs to distinguish the processes generating apparently similar output profiles. Two universities with comparable patent volumes may have different knowledge sources, different combinations of technical elements, and different patterns of subsequent use. Those differences are obscured when all dimensions are reduced to a single performance score.

Information science provides a foundation for examining this problem. Bibliographic coupling, co-citation analysis, and knowledge-domain visualization relate documents through their intellectual connections (Börner et al., 2003; Kessler, 1963; Price, 1965; Small, 1973). Science mapping makes these relations visible across fields and over time (Chen, 2006; Rafols et al., 2010; van Eck & Waltman, 2010). Patent mapping already extends this tradition to technological information, including classification-based overlays and measures of technological distance (Kay et al., 2014; Leydesdorff et al., 2014; Yan & Luo, 2017). Accordingly, the present study does not claim to introduce patents into science mapping. It asks how existing mapping capabilities can be organized into a coherent, temporally explicit account of institution-level knowledge evolution.

Three difficulties motivate the framework. First, relations derived from classification, text, and citations have different meanings. Shared classifications indicate technical proximity, similar language indicates semantic proximity, and a citation records a documented connection to prior art. Treating them as interchangeable links weakens interpretation. Second, evolutionary constructs operate on different timescales. Recombination can be observed when a patent becomes public, whereas recognition, diffusion, and sustained transformation require subsequent observations. Ignoring that distinction can introduce future information into a supposedly early indicator. Third, a cluster of universities at one time is not itself an evolutionary trajectory. A trajectory requires an ordered sequence of observations and evidence that changes exceed classification noise and fluctuations in patent volume.

Chinese university patent data provide the intended empirical setting. A broad institutional population can support comparisons among different technological portfolios without defining the sample through a ranking or restricting it to a small group of highly visible institutions. However, the claim of comprehensive coverage must be established against an explicit university roster and database retrieval audit. University-held patents also delimit the observable knowledge system: without external citing records, the analysis cannot describe economy-wide diffusion. These boundaries form part of the research design rather than assumptions to be hidden in the results.

The study addresses three connected research questions:

- RQ1.** How can acquisition, recombination, emergence, and diffusion be operationalized from complementary patent relations while preserving their conceptual and temporal differences?
- RQ2.** What recurring university-period knowledge profiles and longitudinal trajectories are supported by the observed data, and how much of their variation is explained by patent volume and technological composition?
- RQ3.** Which state transitions coincide with sustained structural transformation, and do the resulting descriptions remain informative under alternative representations, observation windows, and independent validation?

The proposed contribution has three parts. The theoretical contribution is to distinguish four observable process dimensions from transformation as a longitudinal outcome. This makes it possible to ask whether acquisition and recombination are followed by emergence and diffusion without presuming that every invention follows one sequence. The methodological contribution is a common measurement architecture that retains network-layer meaning, defines observation horizons, and connects probabilistic states to directly measured structural change. The disciplinary contribution is to connect knowledge organization with institutional trajectory analysis: maps become evidence for tracing changing relations rather than only visual summaries of a technological landscape. Empirical support for these contributions must come from the tests below; heterogeneous trajectories cannot be inferred from the framework alone.

Related work

Patents as indicators and information objects

Patent statistics are established indicators of inventive activity, but they are selective records of knowledge production. Some inventions are not patented, patent quality varies, and legal and organizational strategies influence observed records (Griliches, 1990; Nagaoka et al., 2010). Citation counts supply additional information, but their interpretation depends on who adds citations and how prior art is identified. Examiner-generated references limit the use of citations as direct

measures of inventor learning (Alcácer & Gittelman, 2006). Spatial spillover analyses similarly depend on the construction of comparison groups and technological controls (Jaffe et al., 1993; Thompson & Fox-Kean, 2005). These findings support a restrained interpretation: patents document technical claims and relations, not the complete cognitive or economic history of an invention.

An information-object perspective adds analytical structure to that restraint. The same patent can be classified within a technical taxonomy, located in a semantic space, connected to predecessors, and followed through later citations. These representations answer different questions and can disagree. Classification systems reflect organized technical categories (Leydesdorff, 2008), while text can identify similarities that are not well represented by a shared class (Kelly et al., 2021; Sarica et al., 2020; Whalen et al., 2020). The useful contribution of integration is therefore traceable complementarity, rather than an unexamined composite measure.

Science mapping and technological knowledge spaces

Classical document-relation methods establish intellectual proximity through shared references or shared citations (Kessler, 1963; Small, 1973). Later mapping systems support the detection of research fronts and the visualization of domain structure (Börner et al., 2003; Chen, 2006; van Eck & Waltman, 2010). Overlay maps enable the comparison of organizations against a common background rather than a separate layout for each portfolio (Rafols et al., 2010). Patent overlays and technological-distance research adapt these ideas to invention records (Kay et al., 2014; Leydesdorff et al., 2014; Yan & Luo, 2017).

Relatedness studies further connect positions in technological space to subsequent entry and exit (Kogler et al., 2013; Rigby, 2015). A map can thus support an evolutionary question, but an observed shift in its layout is not sufficient evidence of transformation. Layout algorithms, changing classification coverage, and changes in sample size can also move points. This study retains a common coordinate system for visual comparison and calculates transformation in the original feature space. It extends an existing mapping tradition through explicit links among relational representation, time-indexed profiles, and longitudinal evidence.

Acquisition, recombination, and technological emergence

Absorptive-capacity theory emphasizes the ability to recognize and use external knowledge (W. M. Cohen & Levinthal, 1990), while exploration and exploitation describe distinct learning orientations (March, 1991). Patent references can provide a partial trace of knowledge sourcing, but cannot establish the entire capability described by these theories. Likewise, technological search recombines components under uncertainty (Fleming, 2001; Fleming & Sorenson, 2004), and search depth and scope need not have the same relationship with innovation outcomes (Katila & Ahuja, 2002). Combinatorial accounts of invention motivate measures of unusual technical combinations (Youn et al., 2015); studies of atypical combinations and patent novelty show why novelty deserves separate measurement from

subsequent impact (Uzzi et al., 2013; Verhoeven et al., 2016; Wang et al., 2017). Emergence is a more demanding construct than novelty alone. Research on emerging technologies emphasizes novelty, growth, coherence, impact, and uncertainty (Rotolo et al., 2015); scientometric methods operationalize selected signals of emerging topics (Small et al., 2014). The present novelty-growth-recognition measure is consequently a bounded operational definition of *recognized emergence*. It does not exhaust the theoretical concept, and its recognition component makes it partly retrospective. Coherence is treated as a representation-validity check, while uncertainty is documented through estimation and coverage diagnostics.

Diffusion, temporal networks, and trajectory inference

Diffusion requires attention to who uses knowledge, how connections extend, and when uptake occurs. Work distinguishing the time to a first citation from later citation accumulation illustrates why a single total discards temporal information (Huang et al., 2019). Here, breadth, depth, and velocity are measured separately before forming a summary diffusion score. They describe documented reuse opportunities, without assuming that every citation is a realized transfer.

Multilayer and temporal network research provides methods for retaining heterogeneous relations and following structural change (Boccaletti et al., 2014; Kivela et al., 2014; Mucha et al., 2010). Community evolution concerns groups that persist, merge, split, or disappear (Palla et al., 2007), whereas the states in this study summarize institutions' measured profiles. Gaussian mixture models represent continuous profile heterogeneity using probabilistic components (Fraley & Raftery, 2002; Scrucca et al., 2016). Those components are useful only if their stability, uncertainty, and temporal interpretation are examined. A Markov matrix then summarizes transitions; it does not establish that a memoryless law governs knowledge evolution.

Technological knowledge evolution framework

Objects, relations, and processes

We define technological knowledge evolution as a change over time in the sources, combinations, recognized directions, and documented subsequent use of technical knowledge, together with sustained changes in its organization. A patent is an observable knowledge object within that process. An organization is a context in which objects accumulate and are connected, rather than a permanent knowledge type.

The framework retains the process vocabulary developed in the earlier draft: *knowledge acquisition*, *knowledge recombination*, *knowledge emergence*, *knowledge diffusion*, and *knowledge transformation*. Figure 1 shows their conceptual relationship. The arrows express possible temporal dependencies to investigate. Acquisition can facilitate recombination; an unusual combination may or may not become a growing, recognized direction; diffusion can reinforce or redirect later

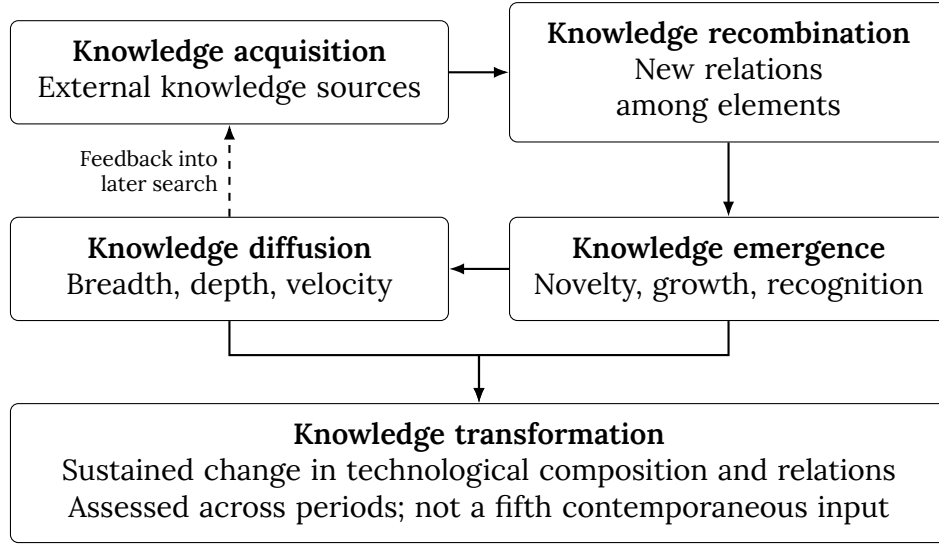


Figure 1. Proposed knowledge evolution framework. Arrows indicate hypothesized relationships, not estimated causal effects or compulsory stages.

search. Internal recombination, unrecognized novelty, and diffusion of established knowledge are all possible. The framework therefore allows branching and feedback instead of prescribing a universal stage sequence.

Four process dimensions and a longitudinal outcome

At university u and period t , the observable process profile is

$$\mathbf{x}_{ut} = (A_{ut}, R_{ut}, E_{ut}, D_{ut})^T. \quad (1)$$

A measures documented external sourcing, R unusual technical combinations, E recognized emergence, and D documented diffusion. Transformation is denoted $T_{u,t \rightarrow t+1}$ and measured separately from the change between periods. The broader framework can be written $\mathcal{K}_{ut} = (\mathbf{x}_{ut}, T_{u,t \rightarrow t+1})$, but the four-dimensional GMM does not include T as an input. This distinction prevents the circular procedure of using future change to construct a state and then claiming that the state explains that same change.

The distinction also clarifies theory testing. Acquisition may increase without recombination; recombination may occur without recognition; diffusion may be strong even when novelty is low. Such profiles are informative counterexamples to a simple cumulative model. The central proposition is that these dimensions carry complementary information about changing knowledge structures. Strong redundancy after controlling for patent volume would weaken that proposition.

From states to trajectories

A *state* summarizes one university-period profile. A *trajectory* is an ordered sequence of profiles, states, and structural changes. We retain four interpretive trajectory categories: *Knowledge Maintenance Path*, *Knowledge Expansion Path*,

Knowledge Recombination Path, and *Knowledge Transformation Path*. Maintenance denotes continuity of a core portfolio; expansion denotes a sustained broadening of coverage; recombination denotes recurrent unusual connections among technical elements; transformation denotes a substantial, persistent reorganization of the portfolio.

These categories are analytic interpretations rather than four classes imposed on the GMM. A selected model may contain fewer or more than four states, and several states may participate in one trajectory category. A university may also follow different paths at different times. In particular, transformation requires direct structural evidence across at least three eligible periods; neither a single cluster switch nor a high citation count establishes it. The terminology does not imply a quality hierarchy: maintaining a coherent specialization can be appropriate to an institution's research mission.

Methods

Patent data as technological knowledge objects

Population, coverage, and analytic unit

The intended source is the research team's Chinese university patent database. Its provider, extraction date, query, and field coverage must be recorded in Table 1; they have not yet been supplied. The candidate retrieval span in the earlier design was 2000–2025. This span remains a proposed extraction range, not a verified observation period. The actual start, end, and mature outcome cohorts are determined by the coverage audit and citation horizon.

The population comprises institutions in a dated, explicitly scoped Chinese university roster and their documented historical names. Territorial and institution-type boundaries are declared with that roster. We retain published Chinese invention applications with a university applicant at first publication, whether or not subsequently granted. Utility models and designs are excluded from the primary analysis because their examination and technical disclosure differ; utility models can be analyzed as a separate sensitivity cohort. Corporate subsidiaries and affiliated hospitals enter only when the affiliation rule and time-specific ownership evidence justify inclusion. Being in the university roster does not guarantee an eligible patent portfolio.

The preferred invention unit is a simple patent family identified by a documented family identifier and its priority set. Publication duplicates and grant versions of the same application are collapsed first. A family is represented by its earliest eligible public invention document, with ties resolved by a stable publication identifier. Later continuations do not contribute text or codes to the earlier representation. Where family identifiers are unavailable, the unit is a deduplicated invention application, and the manuscript must say so rather than claim family-level analysis. A family-level sensitivity analysis is then possible only for the linked subset.

If a unit has m_p distinct university applicants, university attribution uses $\omega_{up} = 1/m_p$ for each applicant university. Thus weights sum to one over universities for the

Coverage item	Value	Required definition or audit
Database/provider; extraction date	TBD	Frozen source and query manifest
Retrieval dates; analytic dates	TBD	First-publication cohorts; mature end date
Roster institutions; linked institutions	TBD	Dated roster and name crosswalk
Raw publication records	TBD	Before publication/application deduplication
Eligible invention units	TBD	Families, or applications if families unavailable
Eligible university-periods	TBD	Three-year non-overlapping cohorts
IPC subclasses; main groups	TBD	Version-harmonized vocabulary sizes
Usable title/abstract records	TBD	Count and coverage percentage
Backward-citation links	TBD	Resolved earlier public invention units
Forward-citation links within H	TBD	Unique family pairs, scope declared
External citing organizations	TBD	University-only or broader coverage
Excluded units by reason	TBD	Mutually exclusive first-failure counts

Table 1. Dataset scale and coverage, to be computed from the frozen source database. No value or date range in this table is an empirical estimate.

university-attributed corpus; this is not a share of legal ownership when firms also co-apply. A sensitivity specification divides by all resolved applicant organizations. We report unique inventions and fractional university totals separately. Organization names are resolved with a dated crosswalk retaining mergers and ambiguous matches for audit.

Dates, windows, and missingness

Each unit is represented by

$$p = (\mathcal{U}_p, \mathcal{C}_p, \mathbf{v}_p, \mathcal{B}_p, \mathcal{F}_p, \tau_p, \alpha_p, \iota_p), \quad (2)$$

where $\mathcal{U}_p$ denotes applicants, $\mathcal{C}_p$ IPC codes, $\mathbf{v}_p$ text features, $\mathcal{B}_p$ backward references, $\mathcal{F}_p$ forward references, τ_p first publication date, α_p application date, and ι_p provenance identifiers. The main temporal clock is first publication, because it represents public availability. Application-year results are a sensitivity analysis, not a substitute for public-information timing. IPC codes are harmonized to a fixed version, with ambiguous concordances reported.

We use non-overlapping three-year publication windows $W_t = [b_t, b_{t+1})$. The preceding three years form the local historical reference window $B_t = [b_t - 3, b_t)$. A five-year forward horizon $H = 5$ years measures diffusion; early recognition uses $h_E = 1$ year. These are proposed protocol settings, not estimated data properties. A cohort contributes to the primary profile only when every included patent has the required follow-up, so $b_{t+1} + H \leq d_{\text{extract}}$ is the conservative period eligibility rule. The earliest cohorts without a complete historical window serve as a burn-in. For the primary clustering cohort, require at least 20 unique inventions per university-period, usable text for at least 80% of its attributed mass, and resolved citation status for at least 80%. Thresholds of 10 and 30 inventions and 70% and 90% completeness are sensitivity settings. A confirmed empty citation list is a zero; an unavailable citation field is missing. Undefined dimensions are not replaced by zero. Missing-dimension and small-portfolio cohorts remain in the coverage analysis and may receive partial descriptive profiles, but are excluded from the primary four-dimensional GMM.

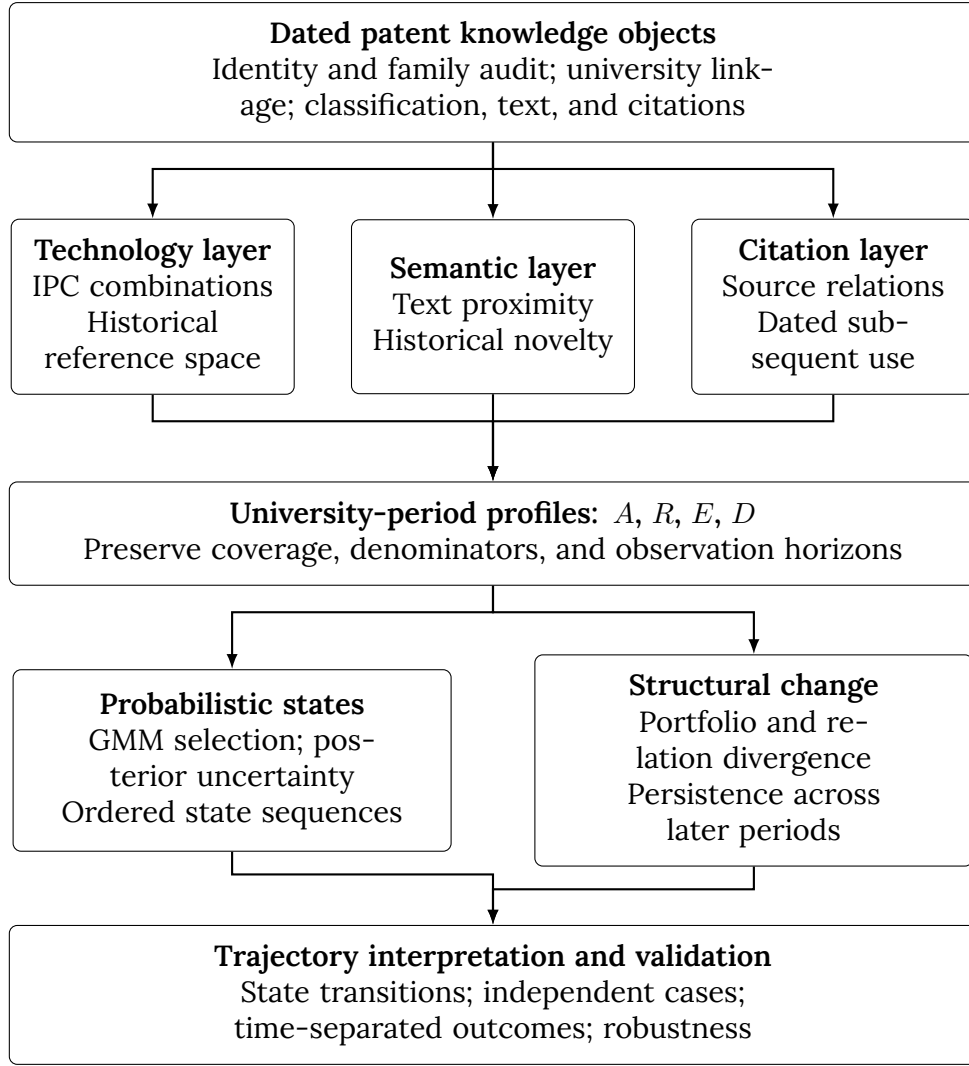


Figure 2. Data-to-inference workflow. Transformational change is measured independently of the contemporaneous state inputs.

Multi-layer knowledge network construction

The representation is a heterogeneous multilayer system, not a multiplex graph with identical nodes in every layer (Boccaletti et al., 2014; Kivela et al., 2014). IPC nodes belong to the technological layer; patent nodes belong to the semantic and citation layers; a patent–IPC incidence relation links their meanings. Figure 2 shows how these sources enter the analysis. No weighted sum of incompatible adjacency matrices is required.

Technology combination layer

IPC subclasses provide the main portfolio vocabulary, while main groups provide finer combination units. For a patent with $m_p^C = |\mathcal{C}_p| \geq 2$ distinct main groups, its

unordered pairs receive equal total mass:

$$w_{ij}^{(t)} = \sum_{p \in W_t: m_p^C \geq 2} \frac{\mathbb{I}\{i, j \in \mathcal{C}_p\}}{\binom{m_p^C}{2}}, \quad i < j. \quad (3)$$

University-specific networks additionally multiply each contribution by ω_{up} . Each multi-coded patent contributes one unit across pairs, reducing inflation from records with many assigned codes. Single-code patents contribute to node mass and portfolio composition, but not to pair weights.

For visualization, normalize with $s_{ij}^{(t)} = w_{ij}^{(t)} / \sqrt{d_i^{(t)} d_j^{(t)}}$, where $d_i^{(t)} = \sum_{j \neq i} w_{ij}^{(t)}$. Edges with a zero denominator are absent. A top-10 incident-neighbor union forms a readable map, with top-5 and top-20 alternatives; indicator calculations use the unpruned weights. Leiden communities are descriptive technical groups (Traag et al., 2019). Map layouts are held fixed across overlays, and neither a community label nor a two-dimensional distance is treated as a causal mechanism.

Semantic layer

The reproducible primary text representation is a character 2–4-gram TF–IDF vector of normalized titles and abstracts. This specification accommodates Chinese technical text without an undisclosed word-segmentation model. Training documents come only from B_t ; retain n-grams present in at least five reference documents and at most 80%, with a maximum vocabulary of 100,000 features. TF is $1 + \log n$ for positive count n , IDF is $\log[(1+N)/(1+\text{df})] + 1$, and vectors are L_2 normalized. Titles and abstracts are separated by a fixed delimiter. Claims are excluded from the primary representation to avoid uneven availability and document-length effects. Empty vectors are missing, not maximally novel.

Semantic similarity is

$$s_{pq} = \frac{\mathbf{v}_p^T \mathbf{v}_q}{\|\mathbf{v}_p\|_2 \|\mathbf{v}_q\|_2}. \quad (4)$$

A mutual 15-nearest-neighbor graph summarizes semantic structure. Retrieval for novelty uses the full eligible historical index, not this pruned display graph. Approximate search is permitted only with a documented recall audit against exact search on a fixed sample. A frozen Chinese-capable sentence encoder is a secondary representation, motivated by embedding approaches (Reimers & Gurevych, 2019; Sarica et al., 2020). Its exact model, tokenizer, revision, training-data limitations, and inference parameters must be recorded before it is used. Modern pretrained representations are treated as retrospective sensitivities unless their historical information boundary can be established.

Citation diffusion layer

Citation edges point from a cited predecessor to a citing descendant, the direction of documented knowledge reuse. Collapse duplicate family-pair citations, remove within-family links, and retain only relations whose predecessor was already public. Same-day or inconsistent-date references are retained in an audit file and

excluded from the time-directed graph. Self-citations are identified by overlap of canonical applicant-organization sets; the primary diffusion measure excludes those edges.

A database containing only university patents supports university-to-university diffusion. A broader claim requires forward citations from all covered organizations, together with their publication dates, identifiers, and IPC fields. Two-step diffusion additionally requires the descendants' outgoing reuse links. Where these are not available, depth is unavailable and the full D score cannot be reported. A two-component breadth-velocity score may be reported as a separately named sensitivity measure, never silently substituted for D .

Knowledge evolution profile extraction

Let P_{ut} be the focal university-period portfolio and $M_{ut} = \sum_{p \in P_{ut}} \omega_{up}$. The following definitions retain natural zeros while distinguishing absent evidence. All normalized components lie in $[0, 1]$; their direction is intensity of an observable process, not research quality. For a dimension j , P_{ut}^j contains only inventions with verified required fields, and $M_{ut}^j = \sum_{p \in P_{ut}^j} \omega_{up}$ is its valid attributed mass. Report M_{ut}^j / M_{ut} for every dimension; missing records never enter an indicator denominator as observed zeros.

Knowledge acquisition: external sourcing and source diversity

For each patent, identify resolved backward-cited inventions whose applicant sets do not overlap its own. Let q_{ut} be the weighted proportion of focal inventions with at least one such external reference. Fractionally allocate each external reference over its IPC subclasses and normalize the aggregate to a source distribution $\pi_{ut,k}$. Let K_t cover the fixed, version-harmonized source subclass vocabulary, including a recorded unresolved-code category; this same vocabulary contains all subclasses receiving source mass. Define

$$q_{ut} = \frac{\sum_{p \in P_{ut}^A} \omega_{up} \mathbb{I}\{|\mathcal{B}_p^{\text{ext}}| > 0\}}{M_{ut}^A}, \quad (5)$$

$$A_{ut} = q_{ut} \left[\frac{1}{2} + \frac{1}{2} \frac{1 - \sum_{k=1}^{K_t} \pi_{ut,k} \log \pi_{ut,k}}{\log K_t} \right]. \quad (6)$$

The entropy term is zero when $K_t \leq 1$. With confirmed absence of external references, $q = A = 0$ and source entropy is recorded as undefined rather than interpreted as a concentrated distribution. The positive baseline inside the bracket distinguishes narrow external sourcing from no sourcing at all. The two components are always reported alongside the composite. This is a documented-sourcing proxy, not a direct measurement of absorptive capacity (W. M. Cohen & Levinthal, 1990; Shannon, 1948; Stirling, 2007).

Knowledge recombination: historical pair rarity

For an unordered IPC-main-group pair (i, j) , let $c_{ij,t}^-$ be its fractional co-occurrence mass in B_t . Let L_t be the fixed vocabulary size established before the focal window, augmented by one out-of-vocabulary category, and $Q_t = \binom{L_t}{2}$. Define the smoothed pair probability and its normalized surprise:

$$p_{ij,t}^- = \frac{c_{ij,t}^- + \alpha}{\sum_{a<b} c_{ab,t}^- + \alpha Q_t}, \quad \alpha = \frac{1}{2}, \quad (7)$$

$$r_{ij,t} = \text{clip}_{[0,1]} \left[\frac{-\log p_{ij,t}^-}{-\log \{\alpha / (\sum_{a<b} c_{ab,t}^- + \alpha Q_t)\}} \right], \quad (8)$$

$$R_{ut} = \frac{\sum_{p \in P_{ut}^{(2)}} \omega_{up} |\mathcal{L}_p|^{-1} \sum_{(i,j) \in \mathcal{L}_p} r_{ij,t}}{\sum_{p \in P_{ut}^{(2)}} \omega_{up}}. \quad (9)$$

$\mathcal{L}_p$ is the set of distinct, valid mapped pairs and $P_{ut}^{(2)}$ contains patents with at least one such pair. A window with no historical pair mass is ineligible. Newly introduced classifications are mapped through the fixed concordance; unresolved new codes use the single out-of-vocabulary category, whose prevalence is reported and excluded in sensitivity analysis. Self-pairs created by that mapping are removed. A university with no eligible pair is assigned missing R , and its multi-code share is reported separately.

Rarity is not a guarantee of novelty or usefulness (Fleming, 2001; Verhoeven et al., 2016; Youn et al., 2015). We therefore inspect the distribution of pair support, compare first-seen-pair rates, and audit text examples. A patent-count-preserving permutation of IPC incidence supplies a reference distribution for combinations that can arise from marginal class frequencies.

Knowledge emergence: novelty, growth, and recognition

For patent p , semantic novelty relative to the historical reference corpus is

$$n_{p,t} = 1 - \max_{q \in B_t} s_{pq}, \quad 0 \leq n_{p,t} \leq 1. \quad (10)$$

Near-duplicate members of the same family cannot be historical comparators. Let $\rho_{k,t}$ denote the fractional share of subclass k in the entire observed reference population during W_t , and $\rho_{k,t}^-$ its share in B_t . The term “reference population” means the university corpus unless broader patent coverage is documented. Relative growth is

$$g_{k,t} = \max \left\{ 0, \frac{\rho_{k,t} - \rho_{k,t}^-}{\rho_{k,t} + \rho_{k,t}^- + \varepsilon} \right\}, \quad \varepsilon = 10^{-12}, \quad g_{p,t} = |\mathcal{C}_p^S|^{-1} \sum_{k \in \mathcal{C}_p^S} g_{k,t}, \quad (11)$$

where $\mathcal{C}_p^S$ contains subclasses. Growth based on shares limits the effect of corpus-wide expansion; raw counts and minimum support are still audited. An unseen field can reach high proportional growth with few documents, so results are repeated after requiring at least 10 current-window inventions per field.

Let $z_p = \mathbb{I}\{p \text{ receives an external citation within } h_E\}$. The primary recognized-emergence score is the mean of patent-level products:

$$E_{ut} = \frac{\sum_{p \in P_{ut}^E} \omega_{up} n_{p,t} g_{p,t} z_p}{M_{ut}^E}. \quad (12)$$

This retains the earlier *novelty* $\times$ *growth* $\times$ *recognition* idea while preventing three unrelated patent subsets from generating a high score through a product of aggregate means. The component means, their dependence, and the fraction with a nonzero product are reported. A zero can reflect lack of any one component. The precursor $E_{ut}^{\text{pre}} = \sum \omega_{up} n_{p,t} g_{p,t} / \sum \omega_{up}$ omits later recognition and is evaluated separately for early-use questions. Coherence checks inspect whether high-scoring records share interpretable technical content; the score alone does not establish an emerging technology in the full sense of Rotolo et al. (2015).

Knowledge diffusion: breadth, depth, and velocity

Let $\mathcal{F}_p^H$ be direct external citing units first published within $(\tau_p, \tau_p + H]$. Define organizational breadth b_p as the number of distinct external applicant organizations in this set. Also report citing-field entropy and the external-university share as descriptive breadth components. Let $\mathcal{F}_p^{(2),H}$ contain unique second-step descendants that lie on a time-respecting path within the same horizon; direct descendants and the focal family are excluded. Define depth and velocity by

$$\ell_p = |\mathcal{F}_p^{(2),H}|, \quad v_p = \begin{cases} 1 - \Delta_p/H, & \mathcal{F}_p^H \neq \emptyset, \\ 0, & \mathcal{F}_p^H = \emptyset, \end{cases} \quad (13)$$

where Δ_p is the time in years to first external citation. Citation arrival at the horizon has zero velocity; zero-citation records remain in the denominator. Depth reflects bounded network reach, not an identified causal chain of learning. Time-to-first-citation and later accumulation warrant separate interpretation (Huang et al., 2019).

For $z \in \{b, \ell, v\}$, use a zero-preserving field-cohort transform:

$$\phi_{f,t}^z(x) = \begin{cases} 0, & x = 0, \\ \{\#(0 < z_q < x) + \frac{1}{2}\#(z_q = x)\} / N_{f,t}^{z,+}, & x > 0. \end{cases} \quad (14)$$

The comparison set contains positive observations in the same primary IPC subclass and three-year cohort. If fewer than 30 positive observations are available, back off to the IPC section and then to the whole cohort; the chosen level is recorded. Field assignment uses the designated primary code, or a deterministic documented code order if that field is absent. If the final comparison set has no positives, all verified observations have value zero. Then

$$D_{ut} = \frac{1}{3M_{ut}^D} \sum_{p \in P_{ut}^D} \omega_{up} [\phi_{f,t}^b(b_p) + \phi_{f,t}^\ell(\ell_p) + \phi_{f,t}^v(v_p)]. \quad (15)$$

Equal component weights are transparent design choices. Raw components and alternative weights are retained because percentile normalization can conceal large absolute changes.

Dynamic pattern identification

Gaussian mixture states

The primary GMM is fitted to all eligible university-period profiles after global dimension-wise standardization. Each row has equal weight, so the fitted population is university-periods; we additionally fit weights summing to one per university to assess effects of unequal panel length. Highly active institutions do not receive weight proportional to their patent volume. The model is

$$f(\mathbf{z}_{ut}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{z}_{ut} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \sum_k \pi_k = 1. \quad (16)$$

Evaluate $K = 1, \dots, 8$ using full and diagonal covariance matrices, covariance regularization 10^{-6} , and 50 seeded initializations. Minimize $\text{BIC} = -2 \log \hat{L} + d \log n$ (Schwarz, 1978). For four variables, $d = 15K - 1$ for full covariance and $d = 9K - 1$ for diagonal covariance. Models with a component below 3% of rows are flagged; no such component is accepted without an explicit stability and substantive assessment. BIC is a selection aid under an approximate distributional model, not proof of natural university types (Fraley & Raftery, 2002; Scrucca et al., 2016).

Assign posterior responsibilities γ_{utk} and descriptive labels $\hat{s}_{ut} = \arg \max_k \gamma_{utk}$. Report maximum posterior probability and normalized posterior entropy. A maximum below 0.60 marks an uncertain assignment; this is a reporting threshold, not a deletion rule. Bootstrap the entire university trajectory, refit standardization and the GMM, and align components to the reference solution using minimum-distance centroid matching. If K changes, report that change and compare co-assignment rather than forcing one-to-one labels. Assess adjusted Rand index, clusterwise overlap, and silhouette descriptions (Hennig, 2007; Hubert & Arabie, 1985; Rousseeuw, 1987). Zeros, boundedness, and skewness may violate Gaussian assumptions, so transformed GMMs and k -means are explicit sensitivity analyses.

State transitions and uncertainty

For consecutive eligible periods with no intervening missing window, define

$$n_{ij} = \sum_{u,t} \mathbb{I}\{\hat{s}_{ut} = i, \hat{s}_{u,t+1} = j\}, \quad n_{i+} = \sum_j n_{ij}, \quad \hat{P}_{ij} = \frac{n_{ij}}{n_{i+}}. \quad (17)$$

Rows with no transitions are undefined. Table 4 reports counts, probabilities, and university-bootstrap intervals. The sensitivity statistic $\tilde{n}_{ij} = \sum_{u,t} \gamma_{uti} \gamma_{u,t+1,j}$ describes posterior-weighted co-occurrence under a factorization assumption; it is not an inferred joint transition posterior. Separate estimates for early and late windows assess nonstationarity. Comparison with a second-order model evaluates whether the previous state improves held-out log loss, using a fixed label space. Stationary distributions are not interpreted unless time-homogeneity is credible.

Whole-university resampling preserves dependence among adjacent transitions (Efron, 1979). We use 1,000 replicates for final intervals and record failed fits. When

null transition opportunities are evaluated, shuffle period order within university while retaining each university's state frequencies, and compare persistence and selected pathway counts to that null. Small cells are reported as uncertain, not converted into general developmental laws.

Transformation as sustained structural change

Let $\mathbf{a}_{ut}$ be the fractional IPC-subclass portfolio distribution, and $\mathbf{r}_{ut}$ the normalized distribution of IPC-pair mass, both represented on a shared vocabulary. Define

$$\text{JSD}(\mathbf{p}, \mathbf{q}) = \frac{1}{2} D_{\text{KL}}(\mathbf{p} \parallel \mathbf{m}) + \frac{1}{2} D_{\text{KL}}(\mathbf{q} \parallel \mathbf{m}), \quad \mathbf{m} = \frac{1}{2}(\mathbf{p} + \mathbf{q}), \quad (18)$$

$$T_{u,t \rightarrow t+1} = \frac{\text{JSD}(\mathbf{a}_{ut}, \mathbf{a}_{u,t+1}) + \text{JSD}(\mathbf{r}_{ut}, \mathbf{r}_{u,t+1})}{2 \log 2}. \quad (19)$$

Natural logarithms yield $T \in [0, 1]$ (Lin, 1991). An unavailable pair distribution makes the two-layer T missing; a composition-only divergence is reported separately. Diversity and field distance can provide complementary interpretation, but are not equivalent to transformation (Porter & Rafols, 2009; Stirling, 2007).

A change exceeds finite-portfolio noise when it lies above the 95th percentile of a within-university null obtained by pooling the adjacent periods' patents and permuting their period labels while preserving period sample sizes. Repeat with portfolio-size-matched subsamples to assess output growth effects. Call it *sustained* only when, in the next eligible period, the new profile is closer to period $t+1$ than to period t , using the same two-layer divergence, and remains distinguishable from the old profile under the permutation test. At least three consecutive periods are therefore required. A high T caused mainly by name resolution, classification revision, or a merger is treated as an organizational or measurement event. Claims of a technological paradigm shift require additional substantive evidence beyond this statistic (Dosi, 1982; Funk & Owen-Smith, 2017).

Empirical validation and robustness

Validation addresses construct distinctness, stability, and independent information. First, report pairwise Spearman correlations, zero shares, and coverage dependence for all dimensions and components. Compare each dimension with log patent volume, portfolio entropy, field composition, and period. Second, conduct blinded case assessment of the original patent descriptions, references, and chronology. Third, evaluate whether profiles add information for outcomes that do not reuse the same observation interval.

For temporal evaluation, set a portfolio-level availability time $a_t = b_{t+1} + H$. All profile inputs must be available by a_t . Outcomes count new external citing organizations during $(a_t, a_t + 3]$ that were not observed as users of the focal portfolio by a_t , and separately assess portfolio change in publication windows beginning after a_t . Training outcomes must finish before the test portfolio's availability time. Re-fit text vocabulary, scaling, normalization references, and any GMM within each training split; no future population percentile or final-period model is transferred

backward in time. If enough separated mature windows are unavailable, temporal validation is declared infeasible rather than replaced by a random row split.

A baseline includes patent volume, average age, university effects where identifiable, period, and field composition. An extended count model adds A, R, E, D , using Poisson pseudo-maximum likelihood and university-clustered uncertainty. Report held-out deviance and mean absolute error; for a binary sustained-change outcome report log loss, calibration, and precision–recall performance. To address early monitoring, a separate model uses only information available at the end of W_t , including E^{pre} and age-observed diffusion, and is not presented as the same matured four-dimensional profile. Prediction improvements are associations, not evidence that changing an indicator will cause an outcome.

For case assessment, select central and uncertain trajectories across model states and portfolio-size strata, before reading validation outcomes. Aim for 24 cases and at least two independent technically qualified raters, subject to the number of eligible trajectories. Show dated evidence without model labels, institution ranking, or score summaries. Raters assess source continuity, technical combination, emerging direction, and sustained structural change. Report raw agreement and Cohen’s kappa with intervals for prespecified nominal labels (J. Cohen, 1960); preserve initial ratings and document adjudication. Cases can disconfirm a model interpretation, not merely illustrate it.

Table 5 defines sensitivity families: record unit, attribution, representation, thresholds, clustering, time windows, citation coverage, and indicator components. Report effect changes, assignment agreement, and coverage changes rather than a single “robust” label. Prespecified secondary associations use Benjamini–Hochberg false-discovery-rate adjustment (Benjamini & Hochberg, 1995). Cluster discovery remains exploratory. Appendix A specifies implementation details and the audit record needed to reproduce each table from the source data.

Results

Empirical reporting status. This section contains the complete reporting structure and interpretive logic for the planned analysis. Numerical entries and data-dependent statements cannot be completed until the database is processed. The marked fields below must be replaced by verified outputs; the existence, direction, and strength of a result remain open.

Coverage and technological knowledge landscape

The final analytic population comprises [Pending: number of universities], [Pending: number of unique inventions], and [Pending: number of university-periods] over [Pending: verified mature publication cohorts]. Table 1 separates raw retrieval from invention-level deduplication and profile eligibility. The principal reasons for exclusion are [Pending: ranked exclusion reasons and counts]. Citation and text coverage vary by [Pending: observed field, period, and institution patterns]. This accounting establishes the population to which subsequent descriptions apply. In particular, a predominance

A. Common technological space

Data required: IPC nodes and weighted pairs.

Position: one fixed reference layout.

Node area: fractional invention mass.

Edges: normalized co-occurrence.

Empirical map not yet computed.

B. Period overlays

Data required: period-specific IPC mass.

Reuse panel A coordinates.

Show entry, persistence, and exit.

Report retained mass and omitted nodes.

Empirical overlays not yet computed.

Figure 3. Technological knowledge space: analysis specification awaiting source data. The panels contain no synthetic nodes, invented clusters, or empirical coordinates.

of highly active institutions after eligibility filtering would narrow the scope from Chinese universities generally to universities with sufficiently observed patent activity.

Figure 3 is reserved for the common technological map and time overlays. Its node count, edge count, community count, and retained mass are [Pending: network statistics]. The overlay comparison reports entry, persistence, and exit of technical subclasses, while keeping node coordinates fixed. An increase in the number of visible areas is interpreted alongside portfolio size and rarefied class coverage. It cannot by itself establish diversification, because a larger sample has more opportunities to include infrequent classes. Semantic neighbors provide a second view of the technical content; discrepancies between the two representations identify records requiring substantive inspection.

Distribution and distinctness of evolution dimensions

Table 2 reports the full distribution of the four dimensions, including medians, interquartile ranges, zeros, and valid denominators. The observed rank order of average intensities is [Pending: rank order], but comparisons across dimensions are secondary because each score has a different substantive meaning. More informative are within-dimension variation and the joint profiles. The correlation between recombination and recognized emergence is [Pending: Spearman correlation and interval]; the correlation between emergence and diffusion is [Pending: Spearman correlation and interval]. These associations must be interpreted with the shared recognition component in mind.

After accounting for portfolio size, period, and field composition, the remaining association between the four scores and output volume is [Pending: adjusted associations]. Weak residual distinctions would indicate that the profile largely repackages established size and field effects. Conversely, substantial variation among size- and field-comparable portfolios would support the usefulness of the relational measurements. Neither interpretation is selected in advance. Component-level results also distinguish low emergence due to limited novelty from low emergence due to delayed recognition; equal composite values can otherwise conceal different processes.

Dimension	Valid n	Mean	SD	Median	IQR	Zero %
Acquisition (A)	TBD	TBD	TBD	TBD	TBD	TBD
Recombination (R)	TBD	TBD	TBD	TBD	TBD	TBD
Emergence (E)	TBD	TBD	TBD	TBD	TBD	TBD
Diffusion (D)	TBD	TBD	TBD	TBD	TBD	TBD

Table 2. Distribution of the four process dimensions across eligible university-periods. Each row also requires observed minimum, maximum, missing count, and component summaries in the accompanying output file.

Selected component	Share	$\bar{A}$	$\bar{R}$	$\bar{E}$	$\bar{D}$	Max. post.
S_1	TBD	TBD	TBD	TBD	TBD	TBD
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$S_{\hat{K}}$	TBD	TBD	TBD	TBD	TBD	TBD

Table 3. GMM state characteristics, with the number of rows determined by model selection. Means require university-bootstrap intervals; “Max. post.” is the mean maximum posterior assignment probability.

Probabilistic states and interpreted trajectories

Among the candidate models, the selected specification has [Pending: selected K and covariance structure], with [Pending: BIC difference from the nearest admissible alternative]. Table 3 reports component prevalence and the original-scale profile means. The proportion of assignments below the posterior-confidence threshold is [Pending: uncertain assignment percentage]. Bootstrap agreement is [Pending: ARI and interval], and the frequency with which the selected state count recurs is [Pending: selection frequency]. These quantities determine how strongly the state solution can be interpreted.

Figure 4 combines the observed state profiles with an ordered trajectory view. A maintenance interpretation requires stable technical composition; an expansion interpretation requires persistent coverage gains; a recombination interpretation requires repeated unusual pair formation supported by text; a transformation interpretation additionally requires the structural and persistence tests. The resulting number of supported trajectory categories is [Pending: supported categories and counts]. Where profiles overlap, uncertain or mixed trajectories remain visible. A state is not renamed “transformation” merely because it has the largest values on all four dimensions.

A. Evolution profile space

Data required: A , R , E , D and posteriors.

Show original-scale state profiles.
Add university-bootstrap intervals.
Use PCA only for a secondary view.

No state centroids estimated.

B. University trajectories

Data required: consecutive period states.

Horizontal axis: publication window.
Rows: prespecified sampled universities.
Mark uncertain and missing periods.

No trajectories assigned.

Figure 4. Evolution profile and trajectory displays: analysis specification. State labels must follow the fitted model, and the trajectory interpretation must use independent structural evidence.

From / to	S_1	$\dots$	$S_{\hat{K}}$	Row count
S_1	TBD	$\dots$	TBD	TBD
$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$S_{\hat{K}}$	TBD	$\dots$	TBD	TBD

Table 4. First-order transition matrix among fitted knowledge-profile states. Each populated probability cell must include its count and 95% university-bootstrap interval; only defined rows sum to one.

A. Observed state movement

Data required: transition count matrix.
 Rows: origin state. Columns: next state.
 Display probabilities and row counts.
 Keep low-support cells visible.
No transition rates estimated.

B. Structural transformation

Data required: divergence and null draws.
 Plot T against its size-matched null.
 Mark persistence in the third period.
 Separate merger and coding events.
No transformation events confirmed.

Figure 5. Transition and transformation evidence: analysis specification. The second panel supplies the structural test required for a transformation interpretation of the first.

Transitions and sustained structural transformation

Table 4 reports transitions among the fitted states. The overall same-state persistence rate is [Pending: rate and interval], compared with [Pending: within-university permutation expectation]. The best-supported off-diagonal transitions are [Pending: transitions, counts, probabilities, and intervals]. These estimates apply to universities observed in consecutive eligible periods; they do not include jumps over missing windows. Early- and late-period matrices differ by [Pending: estimated divergence or stability diagnostic], determining whether an aggregate matrix is an adequate summary.

The earlier draft suggested that recombination could serve as an intermediate route toward transformation. The appropriate test counts ordered sequences with recombination evidence followed by independently established sustained structural change, and compares them with frequency-preserving shuffled sequences. The observed contrast is [Pending: effect size and interval]. A nonpositive or unstable contrast would leave that proposed mechanism unsupported. Even a positive contrast would establish temporal association rather than causation. Figure 5 places transition frequencies beside structural change and persistence so that state switching cannot be mistaken for transformation.

Independent validation, cases, and robustness

In the temporal evaluation, the extended profile model changes held-out deviance by [Pending: change and uncertainty] relative to the volume-field baseline, with [Pending: calibration and absolute-error results]. The mature retrospective model and the early-information model are reported separately. A gain present only when later citation information is available would support retrospective description but

Test family	Alternative and comparison	Observed result
Invention unit	Application versus family; overlap and profile change	TBD
Attribution	University-fractional versus all-applicant and full counts	TBD
Representation	IPC resolution; TF-IDF versus frozen encoder	TBD
Clustering	GMM covariance, transformed GMM, k -means; selected K and ARI	TBD
Window length	Non-overlapping 2-, 3-, and 4-year cohorts; sequence agreement	TBD
Outcome horizon	3 versus 5 years on common mature cohorts	TBD
Eligibility	Minimum 10/20/30 inventions; completeness 70/80/90%	TBD
Citation scope	Self-citations; examiner status; external coverage	TBD
Components	E^{pre} ; separate D components; weight sensitivity	TBD
Independent evidence	Future use; blinded cases; structural persistence	TBD

Table 5. Robustness and validation results, pending computation. Each row requires the common-sample size, effect or agreement estimate, interval where applicable, and change in coverage.

A. Incremental information

Data required: temporally held-out outputs.
Compare volume-field baseline with profile.
Report prediction error and calibration.
Separate mature and early-information models.
No validation gain observed yet.

B. Sensitivity and case agreement

Data required: refits and blinded ratings.
Display effect changes with intervals.
Show state agreement and sample retention.
Include cases that reject model labels.
No robustness claim established.

Figure 6. Validation summary: analysis specification. All comparisons require recorded outcome windows, denominators, and uncertainty; the panels contain no illustrative performance values.

would not justify an early-warning claim.

Independent raters assess [Pending: number of cases]; agreement is [Pending: raw agreement, kappa, and intervals]. The most informative disagreements concern [Pending: evidence-grounded disagreement themes]. Each selected case traces named technical problems, representative invention identifiers, predecessor relations, and later use. A successful explanation must identify what changed in the technology, not simply restate a high score. Cases in which the automated label is rejected are retained to identify construct boundaries.

Table 5 and Figure 6 summarize the effects of alternative assumptions. The largest changes in state membership or transition estimates arise under [Pending: settings and magnitudes]. Stability is assessed on the common eligible sample, while sample changes are reported separately. Any major dependence on a single representation, sparse citation coverage, or the portfolio-size threshold limits the generality of the trajectory interpretation. The final conclusion must therefore be calibrated to the least secure link among coverage, construct validity, assignment stability, and independent evidence.

Discussion

Knowledge evolution as a relational and temporal construct

The principal theoretical value of the framework is the separation of processes that are often compressed into an undifferentiated account of innovation. Acquisition concerns the relation to prior knowledge, recombination concerns relations among technical elements, emergence concerns the alignment of novelty, growth, and recognition, and diffusion concerns later patterns of documented use. Transformation concerns whether these activities are accompanied by sustained changes in the organization of a technological portfolio. Their separation creates testable possibilities for divergence as well as accumulation.

This formulation refines a sequential interpretation of knowledge evolution. An external knowledge source may support routine specialization; an unusual combination may fail to attract recognition; and established technical knowledge may diffuse widely without being novel. The empirical question is whether particular configurations and sequences recur more often than their marginal frequencies predict. If they do, the framework provides an evidence structure for explaining those trajectories. If they do not, its value remains descriptive, and stronger mechanism claims should be withheld. Process vocabulary is useful only when the observable relations remain connected to its theoretical meaning.

Transformation is particularly vulnerable to overstatement. In the present design, portfolio divergence is a measurable form of structural reorganization, not an automatic indicator of a technological revolution. A move into an established field can transform one university's portfolio without changing the global frontier. Conversely, a scientifically important invention may leave the institution's aggregate composition largely unchanged. This distinction locates the contribution at the organization-knowledge-system interface and preserves the broader meaning of technological trajectories in evolutionary innovation theory (Dosi, 1982).

Contribution to information science and knowledge organization

The disciplinary contribution lies in connecting knowledge representation with temporally accountable interpretation. The same invention is linked to a formal classification, a text neighborhood, a set of predecessors, and a later citation neighborhood. Rather than merging these into an opaque similarity score, the framework records which relation supports which inference. A surprising IPC pair supports a combination claim; a remote semantic neighbor supports a content novelty claim; a later citing organization supports a bounded diffusion claim. Disagreement across these views is preserved as a source of analytical evidence. This approach builds on patent mapping, rather than claiming its invention (Kay et al., 2014; Leydesdorff et al., 2014; Yan & Luo, 2017). Its proposed extension is to connect maps, profiles, and trajectories through an explicit unit of observation and a declared time boundary. That connection matters for information analysis because maps are frequently used to support claims about change that their visual construction does not itself establish. Keeping graphical layouts separate from

structural statistics makes those claims inspectable and reproducible.

The framework also treats knowledge organization as historically situated. Classifications change, technical language shifts, and a relation that is unusual at one time may become routine later. Historical reference windows and fixed version mappings provide a controlled basis for studying that evolution, while the audit retains uncertainty introduced by retrospective metadata. The result is a research design for organizing evidence about change, with a clear route from an aggregate description back to the patent records supporting it.

Implications for university innovation governance

The framework suggests a diagnostic use of patent information. A university with strong source breadth but limited recombination poses a different management question from one that repeatedly produces unusual combinations with little subsequent recognition. A stable specialist portfolio raises yet another question: whether continuity reflects productive accumulation or an inability to respond to changing technical opportunities. Such interpretations require substantive context; no score determines the answer by itself.

The four trajectory categories can organize that inquiry without ranking institutions along a single developmental ladder. Maintenance may justify continued support for long-term technical capabilities. Expansion may call for attention to coherence and resource dispersion. Recombination may motivate investigation of the facilities and expertise needed to test unusual technical connections. Transformation may require sustained support and a review of organizational discontinuities. These are conditional questions for governance, not interventions shown to work by the proposed observational analysis.

Any policy-facing use should display coverage, uncertainty, and representative evidence alongside aggregate profiles. A weakly observed university should not be labelled as lacking diffusion merely because external citations are absent from the database. Similarly, a broad profile should not automatically be rated above a specialized one. This restraint is consistent with responsible use of research metrics, where quantitative evidence supports contextual judgment (Hicks et al., 2015).

Limitations and boundary conditions

Several boundaries remain even after the empirical implementation is completed. First, patents cover a selective part of university knowledge production. The framework cannot represent all scientific contributions, tacit expertise, open software, or inventions protected through other means. It does not measure commercialization, licensing success, or market value without additional evidence. Second, citations record formal prior-art relationships, including examiner contributions; they do not directly observe learning or contractual transfer (Alcácer & Gittelman, 2006). These limitations restrict both acquisition and diffusion interpretations.

Third, available data may be narrower than the conceptual network. In particular, university-only forward citations describe a bounded academic technological sub-

system, and missing descendant links can prevent depth measurement. Fourth, novelty is relative to the observed historical corpus and chosen representation. It cannot establish worldwide novelty, nor replace legal examination or technical expertise. Character-based similarity can be sensitive to terminology; pretrained encoders introduce different training and temporal uncertainties.

Fifth, the multiplicative emergence score deliberately penalizes absence of any component and will often be zero. That property makes it interpretable as recognized joint evidence, but unsuitable as the sole measure of early emerging directions. Sixth, the GMM approximates a continuous and possibly non-Gaussian space; unstable or overlapping components may not support categorical summaries. Seventh, a first-order transition matrix can miss longer memory, changing institutional conditions, and cohort effects. Finally, retrospective profiles and subsequent structural changes remain observational. Institutional strategy, staff recruitment, collaboration, and policy changes can affect both. The analysis can identify patterned associations and cases for explanation, but cannot isolate causal mechanisms without additional designs.

Conclusion

Understanding technological knowledge evolution requires a connection between what patent records contain, which relations they document, and when the evidence for change becomes available. This study develops that connection through four process dimensions and a separately evaluated transformation outcome. Its theoretical contribution is the distinction between acquiring knowledge, recombining elements, producing recognized emerging directions, diffusing technical knowledge, and sustaining structural change. That distinction enables the investigation of uneven and branching trajectories rather than assuming that more patent output represents the same form of development everywhere.

For information science, the framework links knowledge organization, relational mapping, and longitudinal inference. Classification, semantic, and citation relations retain their meanings while contributing to a common analysis of institutional knowledge structures. Probabilistic states offer an economical description of profiles; trajectory sequences and direct divergence measures supply the temporal evidence that state labels alone cannot provide. The framework's empirical contribution will depend on whether these distinctions survive coverage checks, size and field controls, independent assessment, and time-separated validation. The present draft establishes the protocol, not those uncomputed empirical findings.

Three extensions follow from the same theoretical architecture. Linking enterprise patent networks would test whether university-originated technical knowledge travels differently across academic and industrial organizations, provided that coverage and applicant histories are comparable. Linking publications to patents would distinguish the acquisition of scientific knowledge from reuse within technological knowledge systems, enabling analysis of science-technology trajectories. Finally, multimodal representations incorporating claims, techni-

cal drawings, and cited passages could clarify what is substantively recombined or transformed, while requiring evidence alignment and uncertainty assessment across modalities. Each extension should improve the interpretation of knowledge change, rather than simply add more variables to an institutional score.

Data, code, and author statements

Data availability. No source patent database or computed empirical output was supplied for this manuscript revision. Database ownership, redistribution conditions, retrieval query, and a permissible reproducibility package must be specified by the authors after the audit. No public dataset deposit is claimed.

Code availability. This package contains manuscript sources, editable vector figures, mathematical specifications, and compilation support. It does not contain an implemented or executed patent-analysis pipeline. A future pipeline should export the provenance and result schema in Appendix B, following traceable data-stewardship principles (Wilkinson et al., 2016).

Author information and support. Author names, affiliations, funding, contributor roles, and competing-interest statements must be supplied and verified by the actual authors. They are not invented for this anonymous working draft.

Appendix A. Reproducible implementation protocol

This appendix supplies operational detail used by the main text. It is a specification for implementation and review, not evidence that the procedures have been executed. All settings should be frozen before inspecting the substantive results, and deviations recorded with their reasons. For a conference-length submission, this extended protocol would need to be separated or shortened in accordance with the applicable venue requirements.

Ordered execution and information boundaries

The analysis proceeds in the following order:

1. Freeze the source extract, provider version, query, university roster, territorial scope, and a checksummed file manifest. Retain the raw source unchanged.
2. Standardize identifiers and dates; collapse publication versions; construct the declared invention unit; identify conflicting family, applicant, and date records.
3. Resolve university and external-organization names with time-stamped mappings. Retain ambiguous matches and the decision evidence instead of silently assigning them.
4. Harmonize IPC codes and create a text-cleaning record. Record the fraction of units affected by classification changes, missing abstracts, and empty vectors.
5. Construct source-scope-specific citation relations; verify external status, date order, and horizon completeness. Determine whether a two-step descendant graph is available before defining the primary diffusion cohort.
6. Establish three-year windows, three-year historical references, and mature follow-up windows. Calculate coverage before imposing the clustering threshold.
7. Build technology, semantic, and citation representations; compute patent-level components; aggregate with the declared university attribution rule.
8. Generate Table 1, Table 2, and all missingness and denominator reports. Investigate invalid ranges before fitting state models.
9. Fit the GMM candidate grid, retain every fit and seed, select the reference solution, and inspect posterior uncertainty. Generate Table 3.
10. Form only consecutive-period transitions. Measure structural divergence and persistence independently. Generate Table 4 and the transition null.

11. Run the prespecified robustness grid on both common and setting-specific samples. Conduct case assessment without revealing model labels to raters.
12. Run temporal evaluation only where availability and outcome boundaries permit. Replace TBD fields with programmatically exported results and review every data-dependent sentence against those outputs.

Minimum source schema

At minimum, the extract must contain publication ID, application ID, patent type, application date, first publication date, applicant names, IPC codes, title, and abstract. Family-based analysis requires a family identifier or an audited priority mapping. A record's citation field requires both its values and a status indicating whether the list is known complete, partially available, or absent.

Each citation relation requires citing and cited identifiers and the citing publication date. Organization-level breadth requires dated applicant identities for citing inventions. Source-diversity analysis requires classifications of cited inventions, even when those lie outside the focal university corpus. Depth requires the next step of citing descendants. If these fields are not available, the affected measures are explicitly unavailable. The manuscript's scope must follow the accessible graph, not the intended graph.

An audit table must map every derived unit to its raw record IDs and record why a representative publication was chosen. Name linkage must store the raw name, canonical organization ID, validity interval, linkage method, decision confidence, and manual adjudication where applicable. Date parsing failures and inconsistent publication order are separate exclusion reasons.

Reference populations and semantic comparison

The focal university portfolio, the source-reference population, and the citing population must not be conflated. The primary historical text and co-occurrence reference population is the observed university patent corpus. If a national all-applicant corpus is available, repeat the novelty and growth calculations against it as a clearly distinct reference specification. A finding of novelty within university patents cannot be relabelled national technological novelty.

Within each semantic fit, apply Unicode normalization, normalize whitespace, remove markup and exact boilerplate only through a recorded rule, and retain technical digits and symbols. Fit the TF-IDF vocabulary and IDF on the historical reference set alone. Store vocabulary checksums and training document identifiers. For patents in a focal period, record the nearest historical neighbor identifier, similarity, and availability date so novelty can be substantively inspected.

For approximate search, sample at least 1,000 focal records or all records when the cohort is smaller. Report top-1 recall and similarity error against exact search. The default acceptance target is recall of at least 0.95; failed recall requires increasing search effort or using exact search, not treating search error as novelty. This is a computational accuracy criterion, not an empirical finding. Sensitivity to title-only text and title-abstract text assesses unequal abstract coverage.

Zero, missing, and undefined values

The primary distinction is evidential. A patent with a verified complete citation list and no external citing unit has zero external diffusion within the observed horizon. A patent with no supplied citation field has unknown diffusion. A university whose patents have one code each has an observable single-code portfolio but no pair-based recombination estimate. A university with no patents in a window has no patent profile for that period; it is not assigned a low-valued profile. These rules prevent data availability from becoming an invented developmental state.

When aggregating partially observed components, the denominator is the eligible attributed mass for that component, and its ratio to total attributed mass is stored. The primary four-dimensional cohort requires the declared completeness threshold and no missing dimension. A secondary missingness analysis compares excluded institutions with retained institutions using observable roster and portfolio characteristics. No claim of unbiased generalization follows merely from a high overall completion percentage.

Cluster interpretation and trajectory coding

Model states initially receive neutral identifiers $S_1, \dots, S_{\hat{K}}$. Order them reproducibly by centroid coordinates for display, but match states across refits by their full profiles. Descriptive state names, if later used, must be based on observed component patterns rather than institution prestige.

Trajectory coding uses three consecutive eligible periods. Maintenance requires continuity of the core subclass portfolio and no sustained structural-change event. Expansion requires a positive change in rarefied subclass coverage across at least two successive periods while retaining a majority of baseline core mass. Recombination requires repeated above-median R relative to field- and period-comparable portfolios plus technical evidence that the unusual pairs represent substantive connections. Transformation requires the divergence and persistence rules in Equation (19), with event-level audit. These rules are proposed coding criteria. Their thresholds must be sensitivity tested and must not be adjusted to produce four well-separated categories.

Core subclasses are the smallest set accounting for at least 80% of baseline fractional portfolio mass, with deterministic handling of ties. Rarefied coverage is estimated by 200 draws of the smaller of the compared invention counts, using the same attribution convention and recording the random seed. Transformation can coexist with recombination or expansion evidence. Where a single mutually exclusive label is necessary for agreement statistics, use the hierarchy transformation, recombination, expansion, maintenance; otherwise retain multiple labels and an unresolved category. Report how often the hierarchy determines the outcome. Do not infer a ranking from this coding order.

Temporal validation safeguards

The main profile is explicitly retrospective: E uses one year of recognition and D uses five years of diffusion. A time label attached to the patent cohort does not

make these values available at its end. The earliest conservative availability time for a complete period is its end plus five years.

For each evaluation split, store the test availability date, all training feature-availability dates, all training outcome end dates, and the test outcome interval. Reject the split if any training label ends after the test availability date. Fit normalization references on training data, with the same field back-off rules; do not compute percentile ranks against the held-out population. For unseen universities, replace unidentifiable university fixed effects with the stated pooled or hierarchical baseline, and report this evaluation separately from predicting later periods of previously observed universities.

The future-use outcome counts organizations newly observed after availability, with all pre-availability citing organizations excluded. A future citing organization that already cited another patent of the same focal portfolio is not new. Store both unique-organization counts and raw citation counts. If the future graph omits firms, label the outcome new academic citing organizations. Absence of enough mature windows is a design limitation, not a reason to reuse the same citations as both predictor and validation target.

Null models and resampling

The main uncertainty unit is the university because its periods and transitions are dependent. For national totals of unique inventions, family-level dependence is additionally relevant; joint patents can connect university records. Sensitivity analysis may resample connected groups of universities sharing inventions, or exclude jointly attributed inventions, to assess this cross-university dependence.

The pair-rarity null uses degree-preserving swaps in the patent-IPC bipartite incidence matrix within the historical field-period strata. Swaps preserve patent code counts and class frequencies and reject duplicate incidence edges. Check mixing using repeated chains and pair-frequency diagnostics. The portfolio change null permutes adjacent-period labels among pooled inventions within university while holding the two period sizes fixed. The transition null permutes state order within university, retaining observed state frequencies and the pattern of eligible windows. These nulls address different questions and cannot be substituted for one another.

Appendix B. Output contract and evidence traceability

The following proposed output files connect each result to a deterministic calculation. They are a schema specification; this manuscript package does not claim to include computed versions of them.

Output	Required contents
<code>source_manifest.json</code>	Provider, extraction time, exact query, roster version, source checksums, field availability, and graph scope.
<code>record_audit.parquet</code>	Raw-to-unit mapping, unit rule, applicant mapping, dates, exclusions, ambiguity flags, and source lineage.

Output	Required contents
patent_components.parquet	Unit ID, cohort, historical neighbor, n , g , recognition, citation breadth/depth/velocity, valid flags, and denominators.
university_profiles.parquet	University, period, attributed mass, A , R , E , D , component means, missingness, source scope, and availability time.
table1_coverage.csv	Unique counts, fractional totals, retrieved and mature dates, coverage, and mutually exclusive exclusions.
table2_descriptive.csv	Valid and missing n , mean, SD, median, quartiles, minimum, maximum, zero share, and uncertainty.
gmm_candidates.csv	K , covariance, seed, likelihood, BIC, convergence, component weights, and rejection flags for every fitted candidate.
table3_states.csv	Selected-state centroids in raw units, intervals, row shares, posterior confidence, and bootstrap stability.
state_posteriors.parquet	All component responsibilities, assigned state, uncertainty flag, model identifier, and preprocessing identifier.
table4_transitions.csv	Origin, destination, count, row denominator, probability, interval, period stratum, and missing-period exclusions.
structural_change.parquet	Portfolio and pair divergences, null distribution summaries, persistence checks, core retention, and event audits.
table5_robustness.csv	Setting, common sample, full sample, selected K , assignment agreement, effect change, interval, and interpretation limitation.
validation_splits.json	Feature availability and outcome dates, all train/test identifiers, fit boundaries, and split feasibility flags.
case_ratings.csv	Prespecified case sampling, blinded evidence IDs, initial ratings, agreement statistics, adjudication, and dissent.

Every numerical manuscript entry should be traceable to a field in an output file and a manifest version. Each result paragraph should state its population, comparison, effect or descriptive estimate, and uncertainty where appropriate. For figures, export node/edge or plotting tables alongside the rendered image. Store original-scale data as well as normalized values so external readers can distinguish a change in absolute activity from a change in relative position.

Appendix C. Symbol and interpretation guide

Symbol	Definition and interpretation
u, p, t	University, invention unit, and non-overlapping publication period.
W_t, B_t	Current three-year window and preceding three-year historical reference window.
ω_{up}, M_{ut}	Fractional university attribution and total attributed portfolio mass.
τ_p, α_p	First publication date and application date; publication is the main clock.
H, h_E, a_t	Five-year diffusion horizon, one-year recognition horizon, and period availability time.
C_p, C_p^S	Distinct mapped main-group codes and subclass codes.
$\mathbf{v}_p, s_{pq}$	Historical-fit text vector and cosine similarity.
$q_{ut}, \pi_{ut,k}$	External-source reach and the distribution of externally cited subclasses.
A_{ut}	Documented sourcing intensity incorporating source diversity.
$c_{ij,t}^-, r_{ij,t}$	Historical pair mass and normalized pair surprise.
R_{ut}	Average unusualness of valid within-patent code combinations.
$n_{p,t}, g_{p,t}, z_p$	Semantic novelty, field-share growth, and early external recognition.
$E_{ut}, E_{ut}^{\text{pre}}$	Recognized emergence and its recognition-free precursor.
b_p, ℓ_p, v_p	External-organizational breadth, bounded second-step reach, and first-citation velocity.

Symbol	Definition and interpretation
D_{ut}	Mean of normalized breadth, depth, and velocity components.
$\gamma_{utk}, \hat{s}_{ut}$	GMM posterior responsibility and maximum-posterior state.
$n_{ij}, \hat{P}_{ij}$	Observed transition count and row-normalized transition probability.
$\mathbf{a}_{ut}, \mathbf{r}_{ut}$	Portfolio subclass distribution and normalized pair distribution.
$T_{u,t \rightarrow t+1}$	Two-layer structural divergence; sustained transformation needs a further persistence test.

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